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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 11: Introduction to Environment Impact Assessment (EIA)

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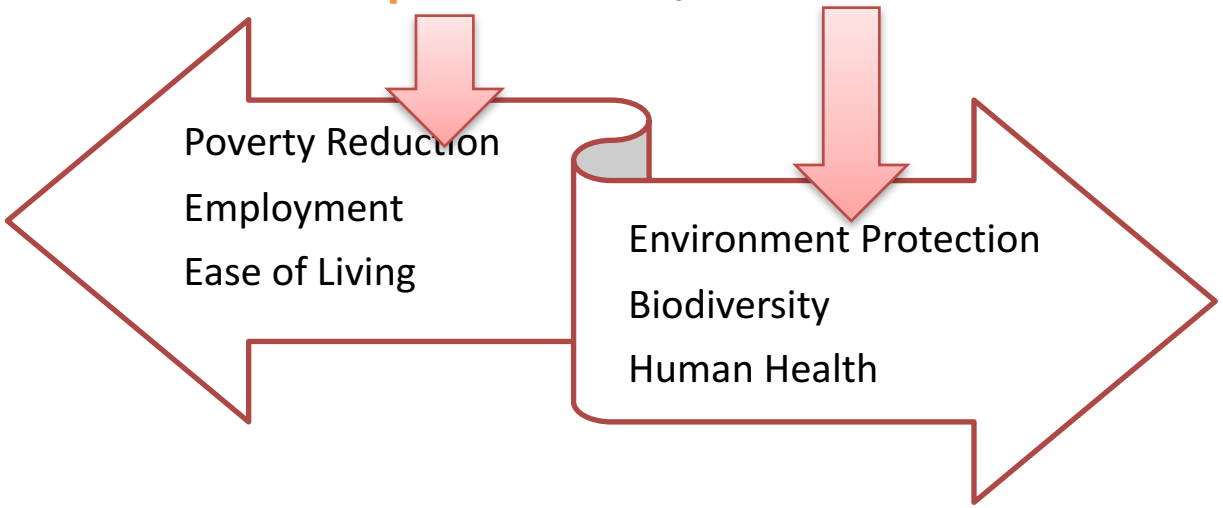


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What will you choose between Development & Environment?

Development V/s Environment



An Everlasting Debate: Development Vs. Environment





Unplanned developments negatively affects the Environment



What is the solution???

Image Source: Delhigreens.com

Development V/s Environment: An Example of the Case of Aarey Forest

- Approx. 800-acre land in Mumbai city
- 2600 trees need to be cut for setting up metro shed.
- Protest erupted after 1500 trees were cut down.
- Later government changed the decision and declared it as reserved forest land.

What should be basis of such decisions?



Image: Aarey Forest
Source: Economic Times

Source: (Financial express)

Development Vs. Environment: An Example of the Case of Kudankulam Power Plant

- The Kudankulam Nuclear Power plant was a project to reduce dependence on coal based thermal power plants in India.
- Villagers protested it citing negative environmental impact.
- An example of a Case of regional/national interest vs. local interests.



Image: Nuclear Power Plant
Source: Indian Express

What should be basis of such decisions?

Source: The Hindu (https://www.thehindu.com/news/national/The-story-of-Kudankulam-From-1988-to-2016/article14564027.ece)

Philosophy of Madhyam-Pratipada

- Buddha suggested: "Enlightenment by avoiding the extremes of self-gratification on one hand and self-mortification on the other" Source: (Britannica)

The Concept of middle path

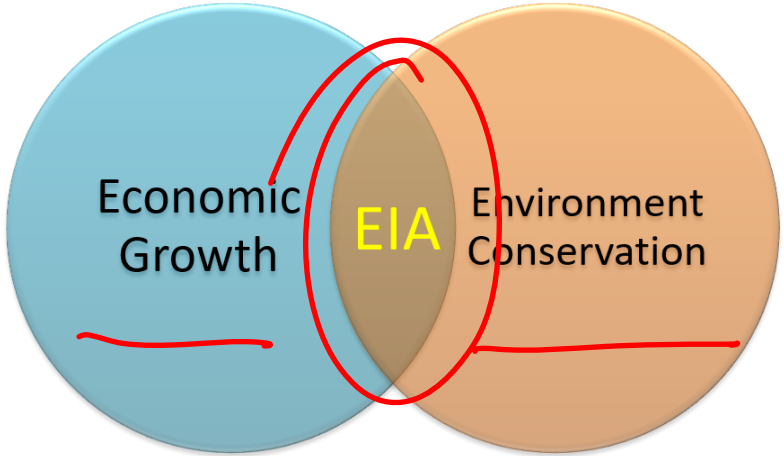
The answer for our debate between development goals vs. environmental goals is by balancing the two sides.



Environmental Impact Assessment (EIA): A Middle Path!



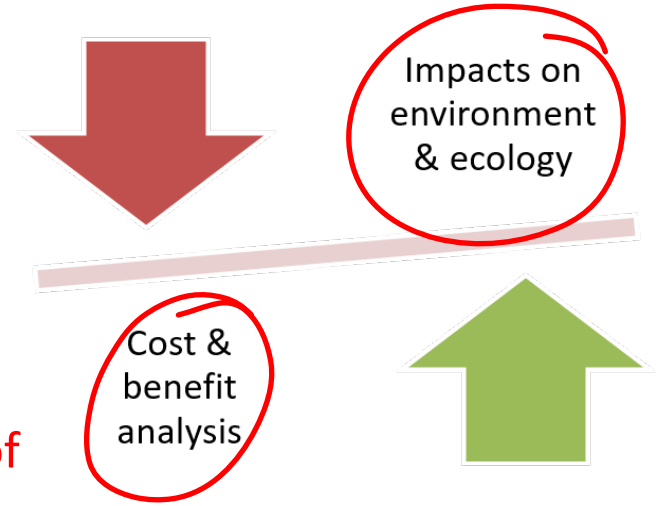
The Solution is EIA



EIA can be termed as balancing process between developmental and environmental goals.

What is Environmental Impact Assessment (EIA)?

- EIA is a tool for decision making to find out the best possible alternative in terms of environmental as well as the economic costs & benefits.

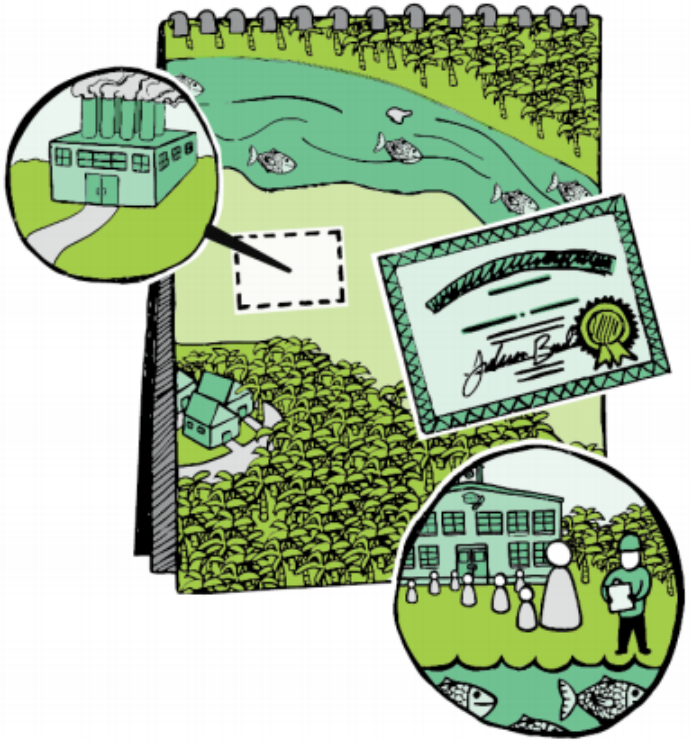


Definition of EIA

Environmental impact assessment (EIA) is a process of identifying, predicting, evaluating, and mitigating the biophysical, social, and other relevant effects of proposed projects and physical activities prior to major decisions and commitments being made.

Source: (Barry Sadler, 1996)

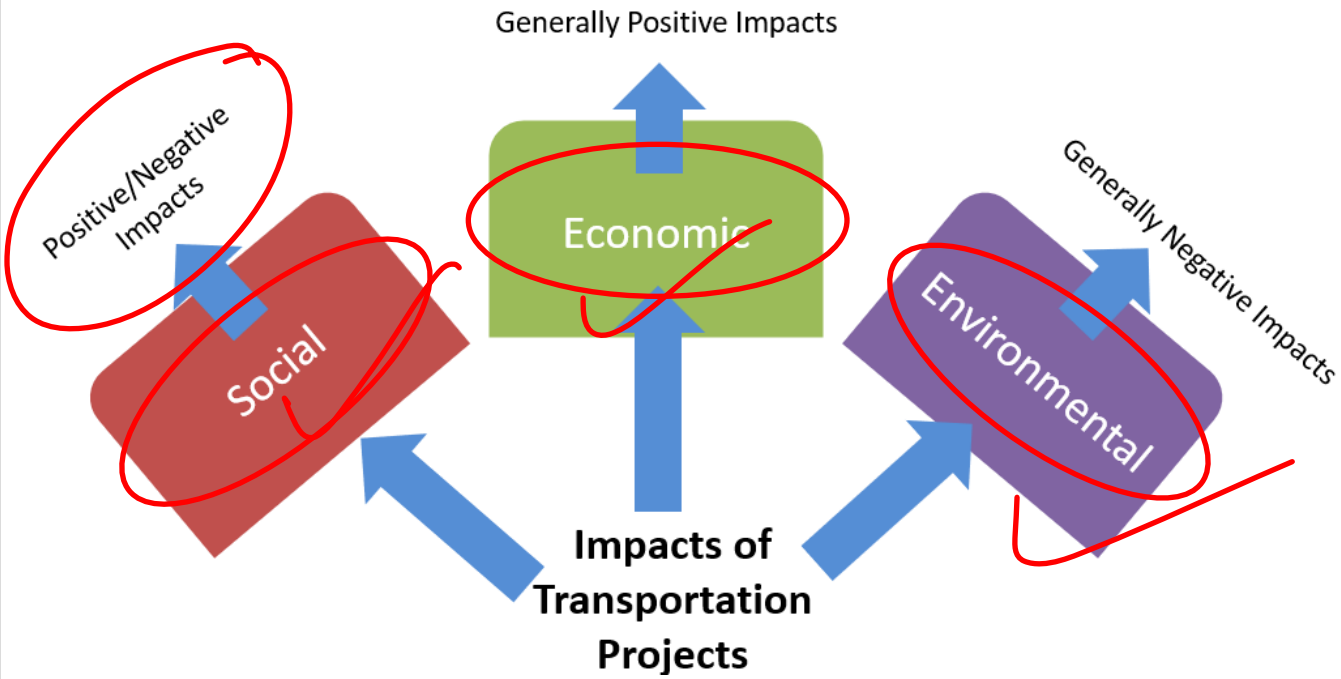
Need for Environmental Impact Assessment Studies



EIA aims to minimize Environmental and social impacts and can enhance Sustainability development in the area, where large-scale projects are implemented.

Source: (IISD, 2016)

Need for Environmental Impact Assessment studies



To consider the Impacts of Transportation Sector in Planning

Purpose of Environment Impact Assessment

To facilitate decision-making

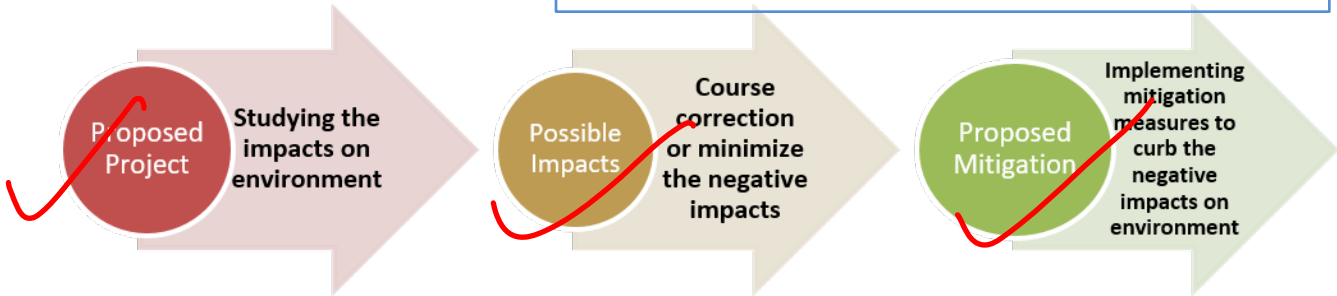
- Systematic examination of the environmental implications of a proposed action, and sometimes alternatives, before a decision is taken.

To aid in the formation of development

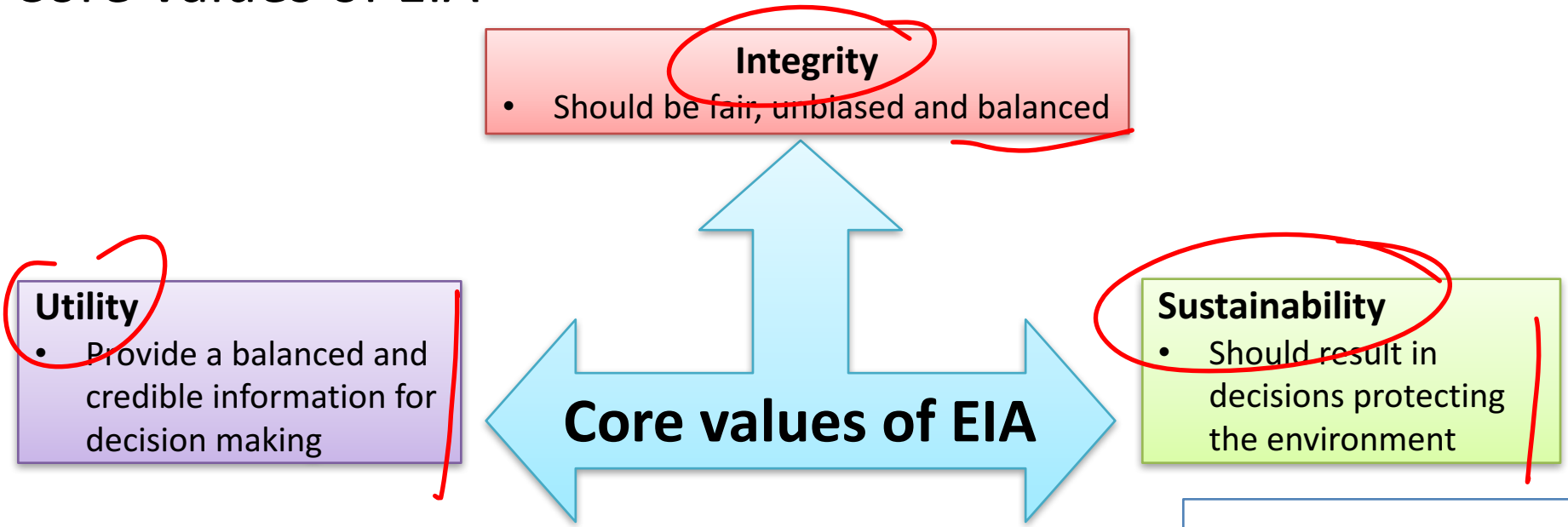
- Framework for considering location, design and environmental issues in parallel.
- Aid to the formulation of developmental actions
- Environmentally sensitive developments

To be an instrument for sustainable development

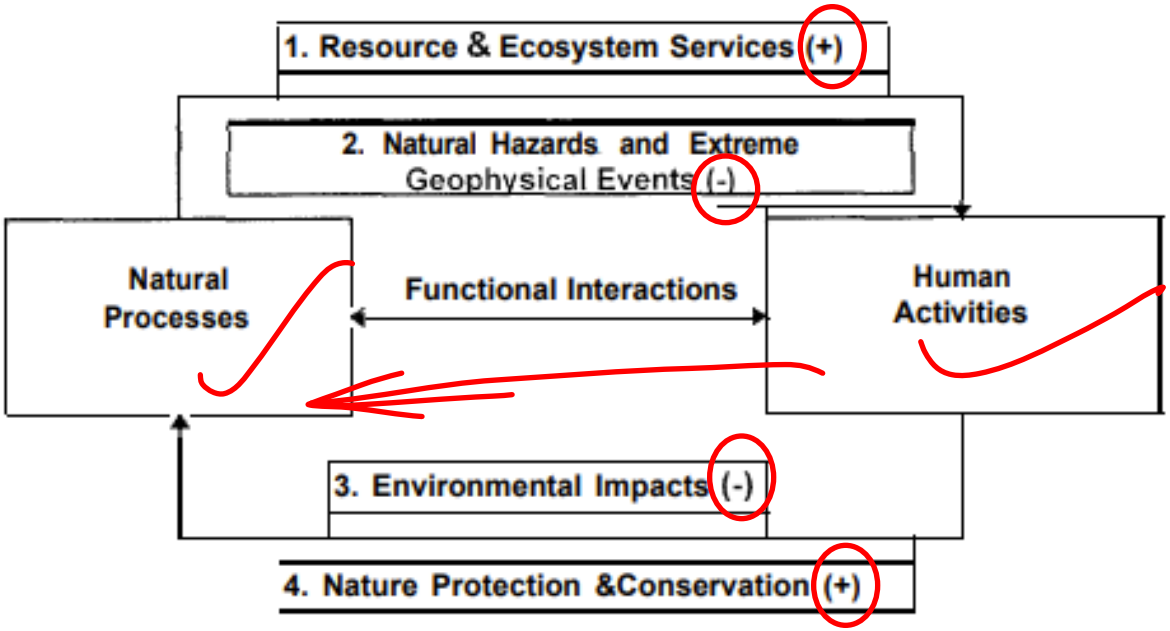
- Overall quality of life
- Continuing access to natural resources
- Avoiding lasting environmental damage



Core Values of EIA

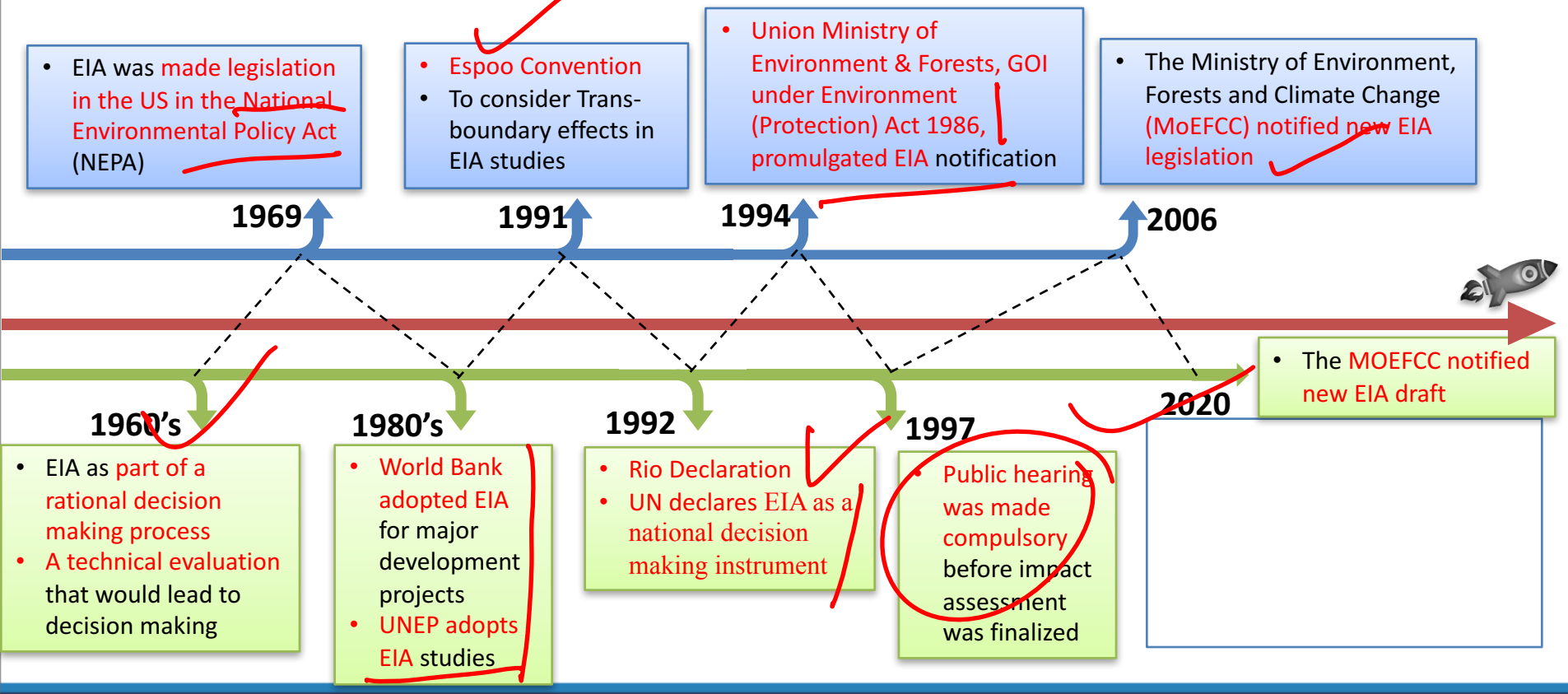


Components of Environment Impact Assessment



1. Resource capability and ecosystem mapping
2. Natural hazard and landscape sensitivity evaluation
3. EIA and other forms of Impact Assessment
4. Ecosystem approaches ex. Critical zone, habitat supply etc.

Timeline of Environmental Impact Assessment



Evolution of EIA in Countries Worldwide

Period	Development of EIA
Pre-1970	• Project review based on the technical/engineering and economic analysis. • Limited consideration given to environmental consequences.
Early/mid – 1970s	• EIA introduced by NEPA in 1969-70 in US. • Standard methodologies for impact analysis developed (e.g. matrix, checklist and network). • Canada, Australia, New Zealand and Columbia became the first countries to follow NEPA in 1973-1974. • Australia legislated EIA, whereas Canada and New Zealand established administrative procedures.

Source: (IAIA, 1996)

Evolution of EIA in Countries Worldwide

Period	Development of EIA
Late 1970 and early 1980s	• More formalized guidance. • Other industrial and developing countries introduced formal EIA requirements • (France, 1976; Philippines, 1977), began to use the process informally or experimentally (Netherlands, 1978) or adopted elements, such as impact statements or reports, as part of development applications for planning permission (UK, 1988). • Use of EIA by developing countries (Brazil, Philippines, China, Indonesia) • Strategic Environment Assessment (SEA), risk analysis included in EIA processes. • Provision for public involvement. • Coordination with land use planning processes.

Source: (IAIA, 1996)

Evolution of EIA in Countries Worldwide

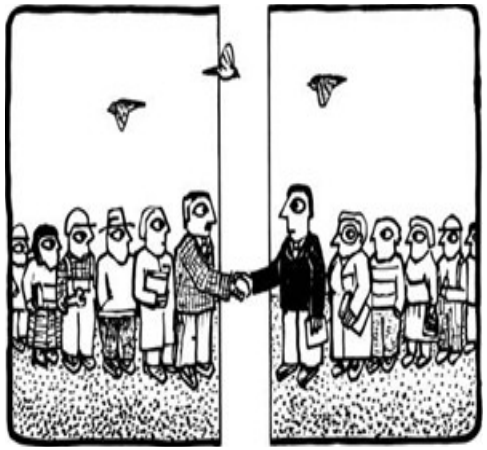
Period	Development of EIA
Mid 1980s to 1990	• In Europe, EC Directive on EIA establishes basic principle and procedural requirements for all member states. Increasing efforts to address cumulative effects. • World Bank and other leading international aid agencies establish EIA requirements. • Spread of EIA process in Asia
1990s	• Requirement to consider trans-boundary effects under Espoo convention. • Increased use of GIS and other information technologies (IT). • Sustainability principle and global issues received increased attention. • India adopted the EIA formally.

Source: (IAIA, 1996)

Evolution of EIA- Key International agreements

Convention on Environmental Impact Assessment in a Trans-boundary Context (Espoo, 1991).

- This is the first multi-lateral EIA treaty.
- Treated EIA in a trans-boundary context and entered into force in 1997.
- The Espoo Convention sets out the obligations of Parties to assess the environmental impact of certain activities at an early stage of planning.
- It also lays down the general obligation of states to notify and consult each other on all major projects under consideration that are likely to have a significant adverse environmental impact across borders.



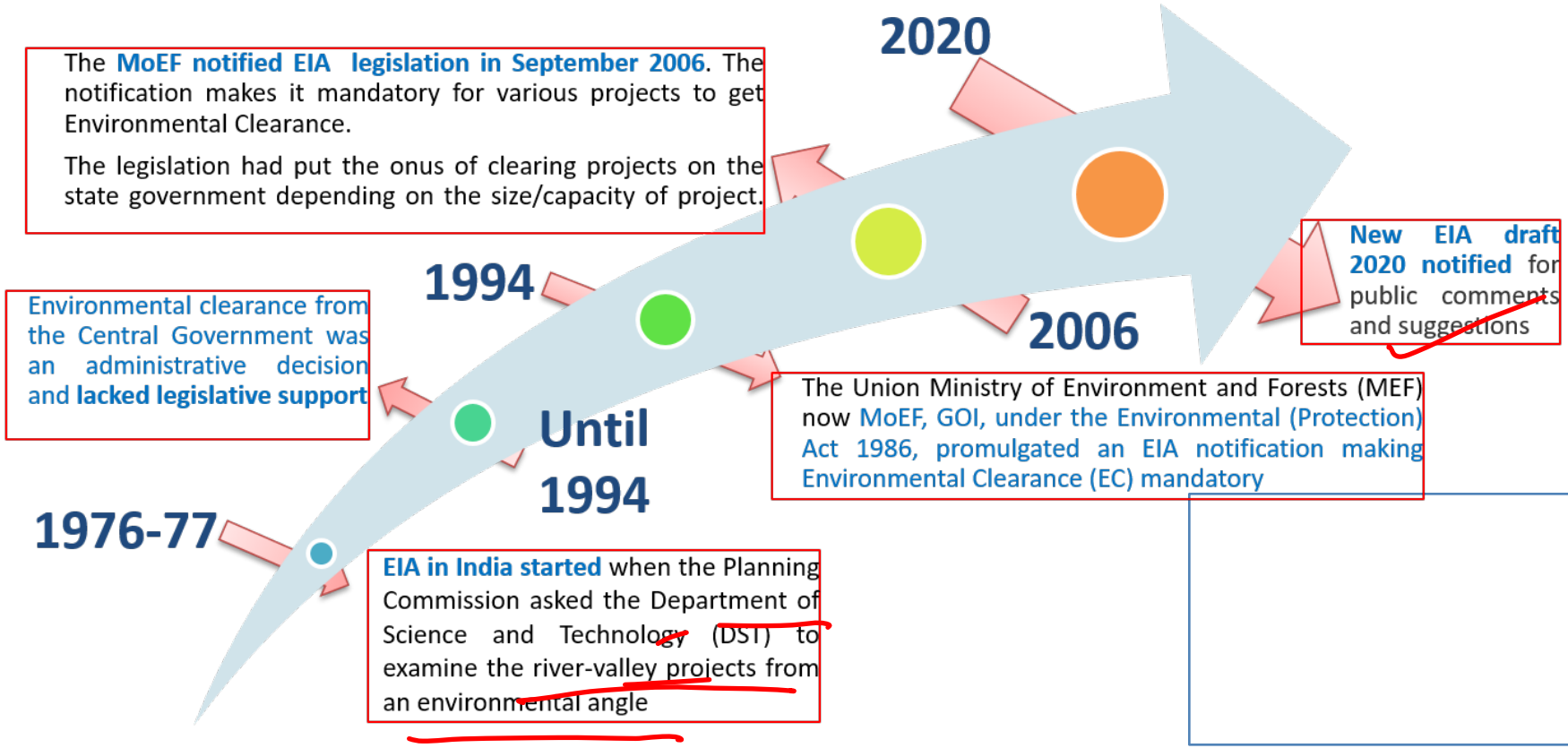
Evolution of EIA- Key International agreements (cont'd..)

Rio Declaration (1992)

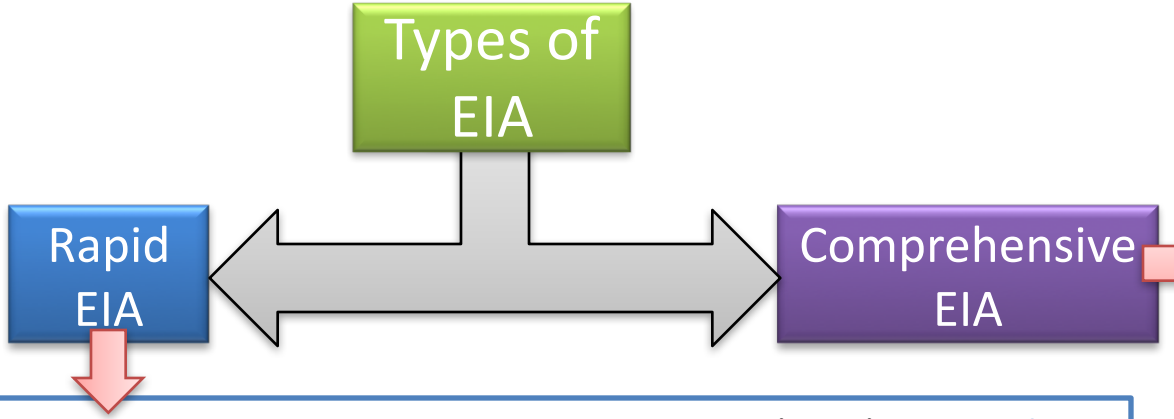
- Principle 17 of Rio Declaration on Environment and Development calls for use of EIA as a national decision making instrument to be used in assessing whether proposed activities are likely to have significant adverse impact on the environment.
- It also emphasized the role of competent national authority in the decision making process



EIA Timeline in India



Types of EIA



- Rapid Environmental Impact Assessment (REIA) is done for quick assessment of likely adverse impacts.
- Mostly conducted at the initiation of the new activity.
- Rapid Environmental Impact Assessment study covers one season baseline data for various environmental components i.e. Air , Noise, water, land, Biological and Socio-economic including parameters of human interest.

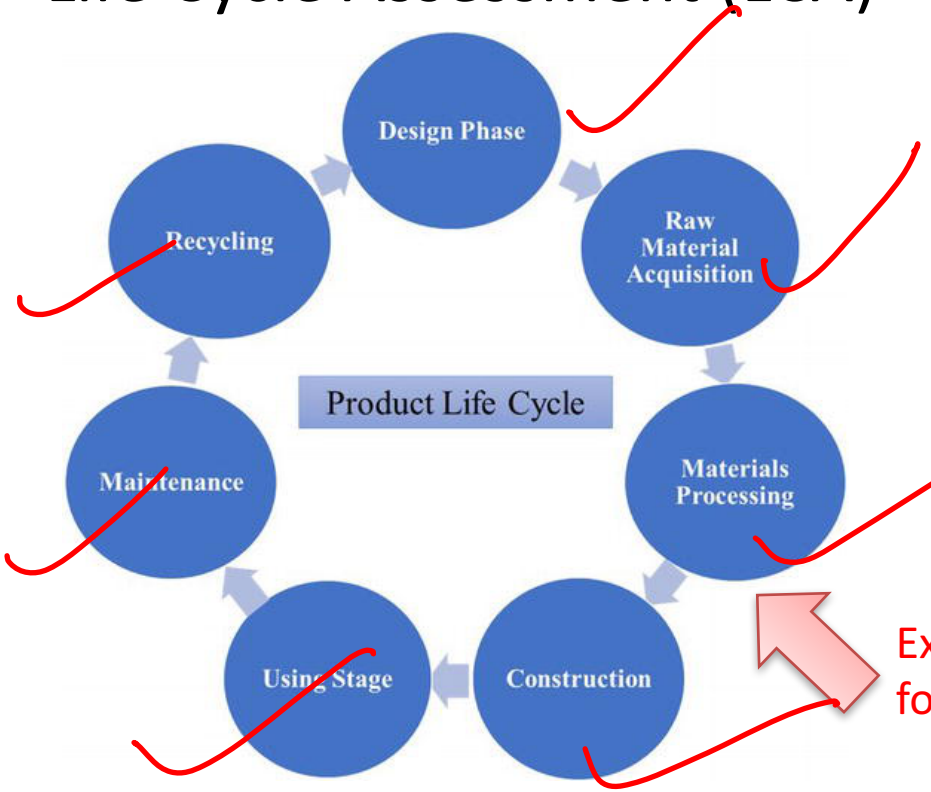
- Comprehensive Environmental Impact Assessment includes collection of data for various components for four seasons i.e. Monsoon, Post-monsoon, Winter and Summer.
- Time taking process
- Conducted only if required and asked by the assessment body.

Objective-based Classification of EIA



EIA can be classified on the basis of objectives of the project or its theme.

Life Cycle Assessment (LCA)



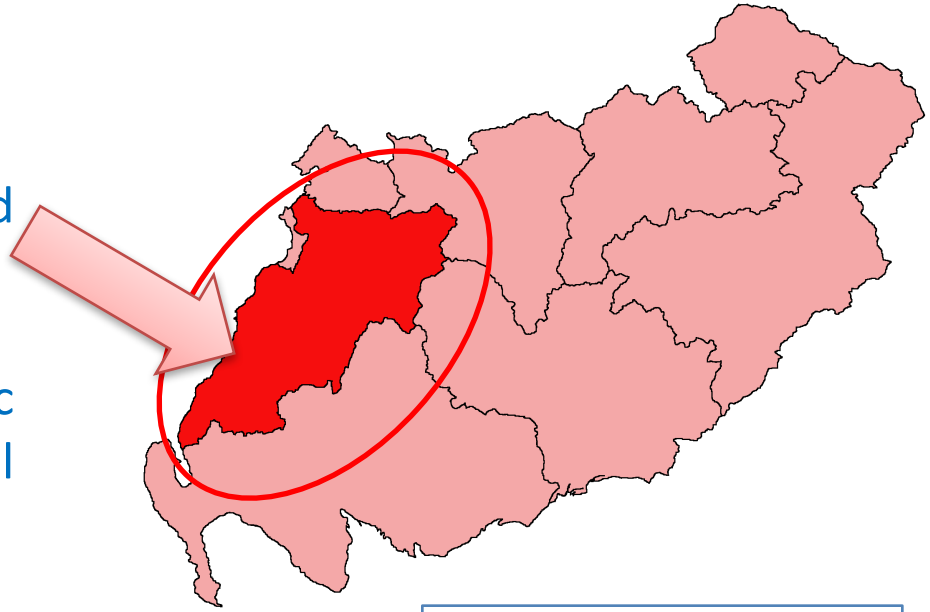
- Highlights that environmental concerns enter every step of the process.
- Examines environmental impacts of the product at all stages of the product life cycle.
- Way to deal with environmental impacts, especially in manufacturing.
- Considers the total picture rather than just one stage of the production.

Example of LCA for a Product

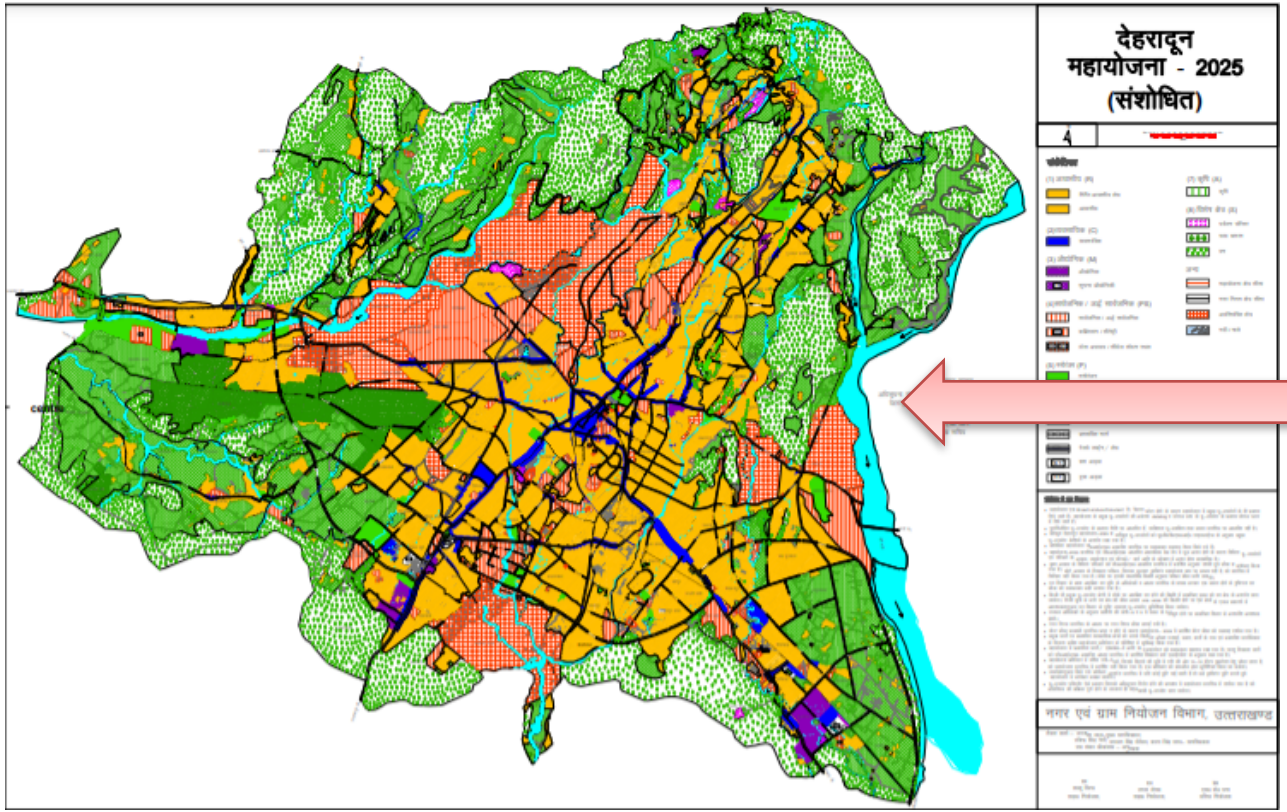
Source: (Jiawen Liu et. al, 2019,)

Regional EIA

- Environmental concerns integrated with regional planning.
- Adequate integration of economic growth with management of natural resources.
- Critical for achieving sustainable development goals.



Regional EIA example of the Doon Valley, Uttarakhand



- Doon valley is situated around the Dehradun (Capital city of Uttarakhand).
- Map showing the master plan of Doon valley submitted by state government

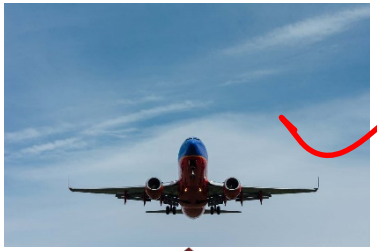
Regional EIA example of the Doon Valley, Uttarakhand (cont'd..)

- Location of industries will be as per Water (Prevention and Control of Pollution) Act, 1974 and the Air (Prevention and Control of Pollution) Act, 1981.
- Land use, tourism and grazing will be as per plan approved by ministry of Environment, Forest and Climate Change (known as MoEF).
- Any mining activities needs prior clearance from MOEF.
- Red category industries (most polluting) is not allowed.



Image:
Doon Valley

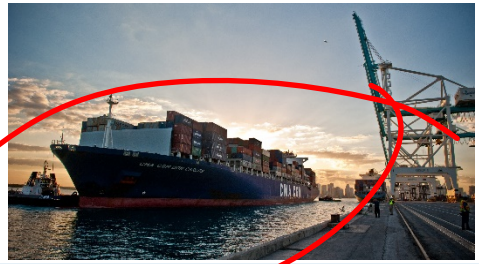
Sectoral EIA





Transportation sector





- EIA concerning specific sectors such as transportation, energy, mining etc.
- Addresses specific issues encountered during implementing any sectoral project/s.

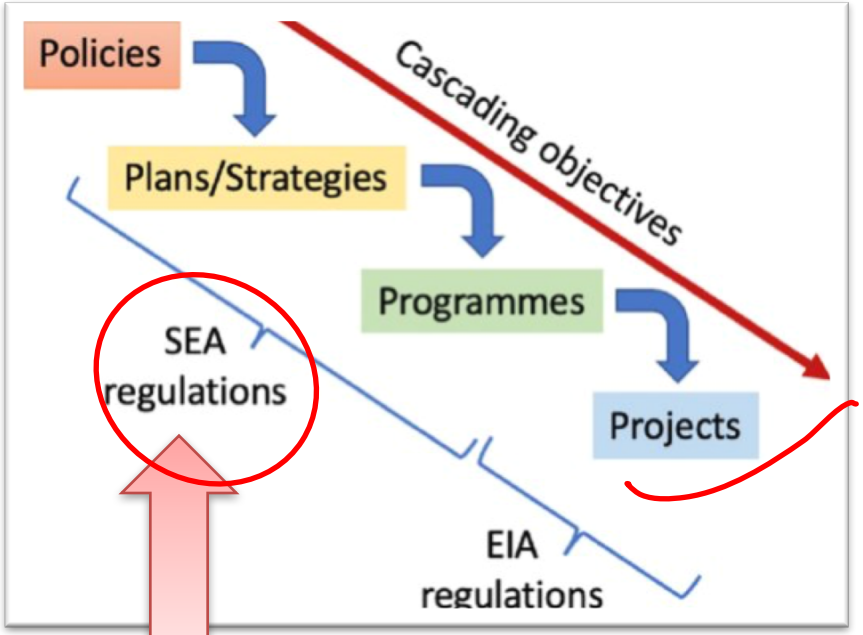
Sectoral EIA for Transportation Sector

Key Features:

- Wider impacts (on national or global level).
- Promotes integration of various transportation modes.
- Overall assessment of energy and fuel consumption by transport sector and implementation of best strategy to minimize adverse impacts.
- Encourage better usage of resources in sustainable manner.



Strategic EIA (SEIA or SEA)



Relationship between Strategic Impact Assessment and Environment Impact Assessment

- SIEA is the systematic analysis of the environmental effects of development policies, plans, programmes (PPP) and other proposed strategic actions including the preparation of a report that helps in decision-making.
- Proactive approach to integrate environmental concerns in the decision-making process.

SEA is the EIA of policies, plans and programs

Source: (Javier Rodrigo-Ilarri et. al, 2020)

Difference between EIA and SEA

Environment impact assessment	Strategic environment assessment
• Takes place at end of decision-making cycle	• Takes place at earlier stages of decision making cycle
• Reactive to a development proposal	• Pro-active to development proposals
• Identifies specific impacts on the environment	• Access effects of a policy, plan or programme on the environment or the effect on environment on development needs and opportunities.
• Addresses a specific project	• Addresses areas, regions or sectors of development
• Considers limited number of feasible alternatives	• Considers broad range of potential alternatives
• Emphasis on mitigating and minimizing impacts	• Emphasis on meeting environmental objectives, maintaining environmental quality
• Narrow perspective and a high level of detail	• Broad perspective and lower level of detail to provide a vision and overall framework
• Well-defined process, clear beginning and end	• Multi-stage process, overlapping components, policy level is continuing and iterative
• Focuses on project specific impacts.	• Creates a framework against which impacts and benefits can be measured.

Example of SEIA for Helsinki Metropolitan Transport System (Finland)

- 3 alternatives were compared
  - Existing 1994 system (Baseline scenario)
  - Based on policy to develop public transport
  - Based on policy to develop private vehicle traffic

- For each alternative, positive and negative points were identified based on:
  - Land use
  - Environment
  - Traffic and mobility



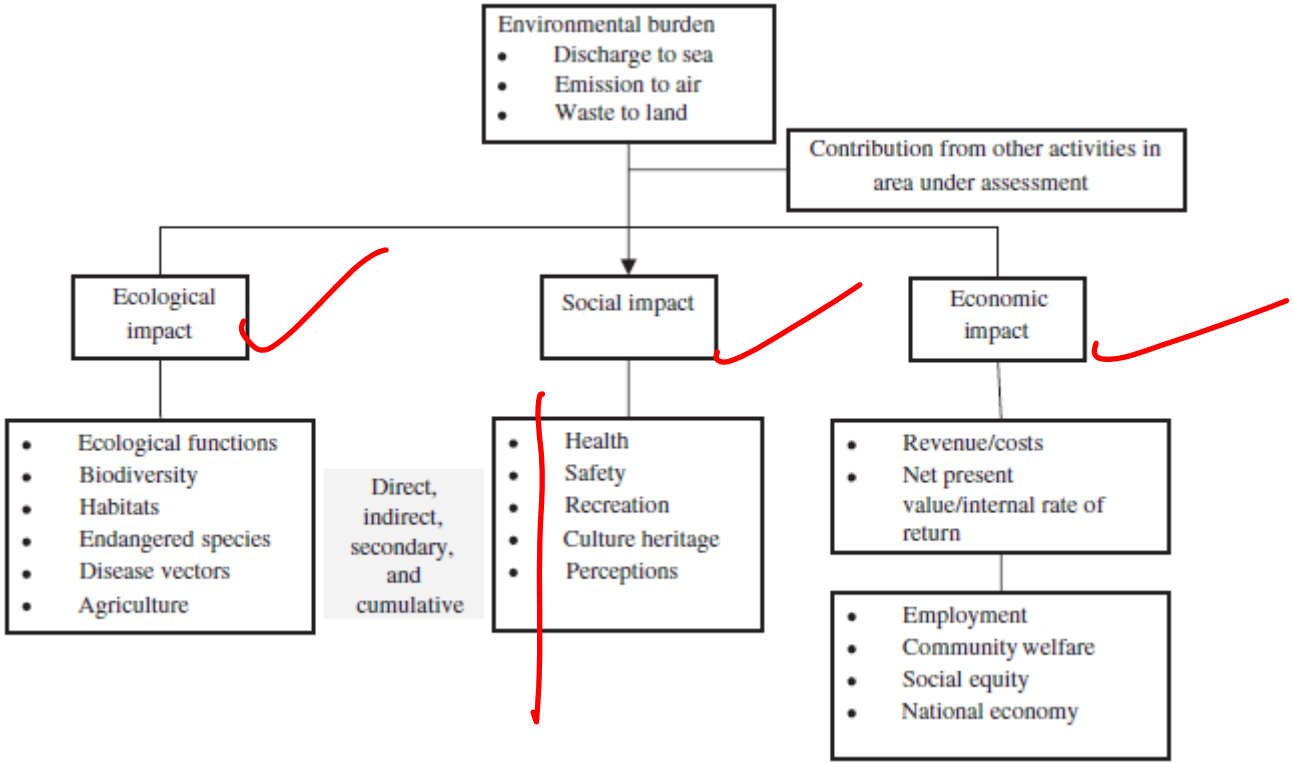
Helsinki Metro

Project level EIA

- Developmental activities considered in isolation.
- Impact assessment specifically for activities proposed in project.
- Most common EIA process
- Example: EIA for any section of highway or a small project.



Classification of Environmental Impacts



Source: (Bai Yong and Bai Qiang, 2014)

Beneficial or Detrimental Impacts



Image Source: nhpr.org

Example: Transport projects such as new construction of roads, widening of roads

- Detrimental impact: Deteriorating air quality
- Beneficial impact: Improved air quality by using roads for public transportation and Non-motorized transports

Naturally Reversible or Irreversible Impacts



Example: Waterbodies such as rivers

- Reversible impact: Maintaining natural flows enhances self purification properties of water bodies (by increased dissolved oxygen levels).
- Irreversible Impact: Extensive pollution leads to extinction of species.

Repairable or Irreparable Impacts

Impacts Repairable via management practices or irreparable

Example: Deforestation for Road construction

- Repairable impact: Forest cover may be increased through plantation.
- Irreparable impact: Irreparable once sensitive ecological cycle is disturbed, it is irreparable.



Short-term or Long-term Impacts



Congestion on roads due to construction activities of a Metro project

Source:(https://www.dnaindia.com/mumbai/)

Example: A Metro Project

- Short-term impact: Congestion on roads during construction period.
- Long-term Impact: GHG emissions during the entire project life of a metro.

Local, Regional, National Scales & Impacts



Local Impacts example of Dirt from Unpaved roads and construction sites



Regional Socio-economic Impacts example of benefits of BRTS or MRTS



Regional Environmental Impacts example of Acid rain due to SO_X / NO_X emissions

National Impacts example of Alternate fuels (Ex. Implementation of BS VI) in India

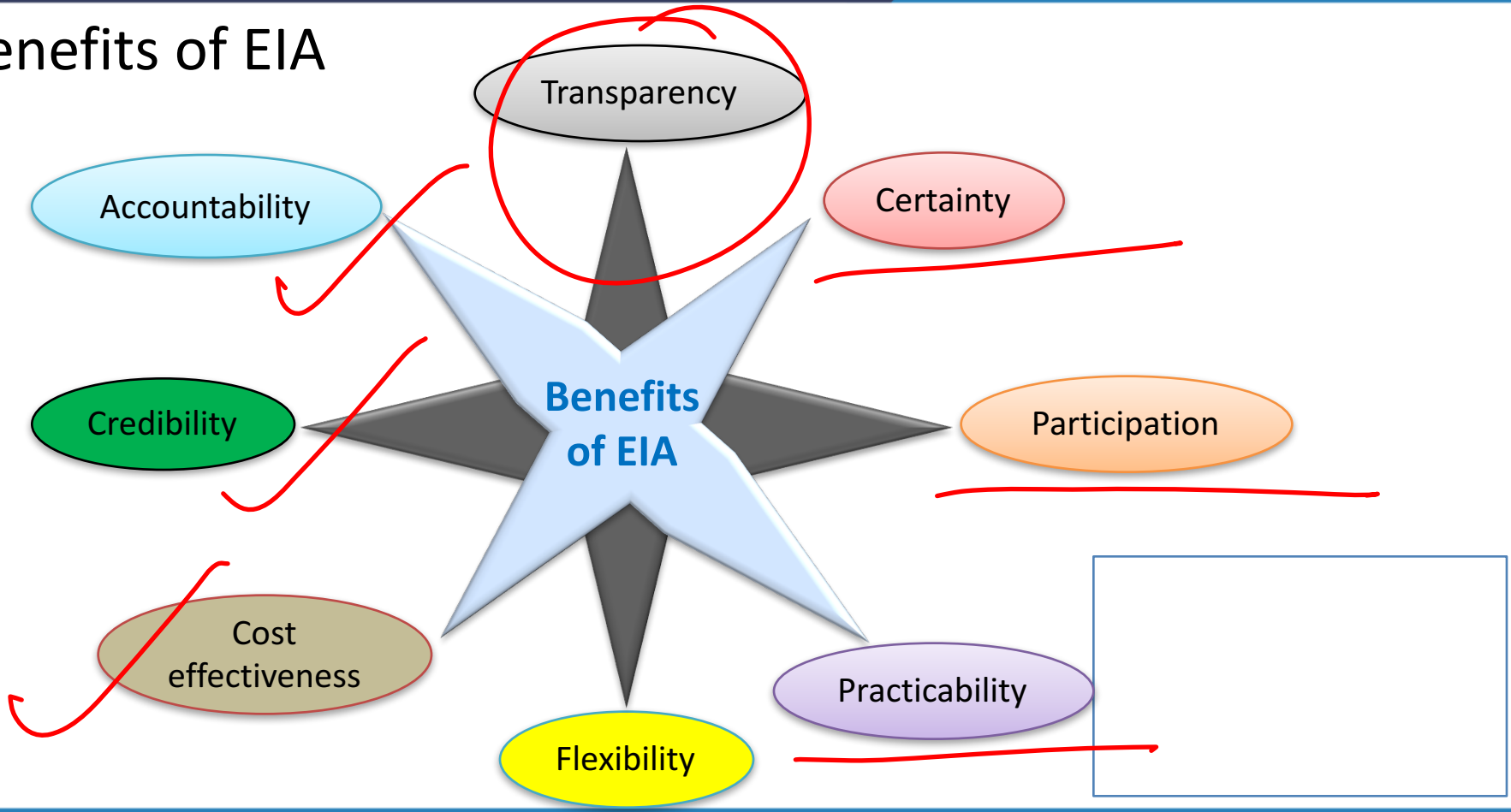


Forms of Impact assessments

- Social Impact Assessment
- Risk Assessment
- Life Cycle Analysis (LCA)
- Energy Analysis
- Health Impact Assessment
- Regulatory Impact Assessment
- Species Impact Assessment
- Technology Assessment
- Economic Assessment
- Cumulative Impact Assessment
- Strategic Environmental Assessment (SEA)
- Integrated Impact Assessment

Based on the type of Project and the need for study, an EIA study can include any of the following Impact assessments in the study.

Benefits of EIA



Benefits of EIA and their Levels of Success

Benefit of EIA	Very successful	Moderately successful	Marginally successful	Not Successfu
Including a full range of considerations (e.g., social, ecological, risk, etc.)	11%	53%	27%	6%
Making precise, verifiable predictions.	2%	34%	45%	15%
Identifying appropriate mitigation measures.	12%	57%	24%	3%
Indicating confidence levels for data used in predicting impacts.	3%	17%	43%	32%
Specifying the significance of residual impacts.	3%	28%	45%	19%
Providing clear, understandable information for decision makers on the potential consequences of development proposals.	14%	51%	26%	8%
Providing relevant advice to decision makers on alternatives to the proposal being assessed.	8%	34%	37%	16%

Source: Sadler, 1996.

According to International Institute of Sustainable Development (IISD), 2016.

Source: (IISD, 2016)

Comparison of Benefits and Flaws in EIA Studies

Benefits	Flaws
• Provides systematic method of impact assessment	• Time consuming
• Estimates the Cost/Benefit trade-off of alternate actions	• Costly
• Facilitates the public participation	• Little public participation in actual implementation
• Provides an effective mechanism for: • Coordination • Environmental Mitigation • Negotiations • Feed back	• Unavailability for reliable data (mostly in developing countries
• Top-level decision making	• Too focused to scientific analysis (sometimes)
• Achieves a balance between the impact of developments and environmental concerns	• Compliance monitoring after EIA is seldom carried out

Source: (School of Planning and Architecture)

Positive and Negative Externalities

+ve

Positive Externalities

1. New jobs generated, economic growth is stimulated.
2. Growth of local business enterprises supported.
3. Development of supporting and complementary industries.
4. Influx of capital and disposable income.

Negative Externalities

1. Additional pressure on Physical Infrastructure
2. Escalation of Cost
3. Extended time in project implementation

-ve

Conclusion

- Environmental Impact Assessment (EIA) is an effort to having harmony with nature while without compromising the economic benefits of growth and development activities.

- EIA considers various environmental impacts with respect to its type, level or nature.

- An EIA study not only prevents any damage to natural resources and environment, but also helps to gain developmental and economic profits in long-term, thereby ensuring Sustainable development.



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Thank You